Contact

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www.linkedin.com/in/tanmay-bhuskute (LinkedIn)
tanmayb05.github.io/portfolio/ (Personal)

Top Skills

Recommender Systems
k-means clustering
K-Nearest Neighbors (KNN)

Certifications

Introduction to Programming Using Python

Tanmay Bhuskute

Software Engineer | Ex-Siemens | M.S. Computer Science @ University of Cincinnati | Cloud • AI • Distributed Systems
Cincinnati, Ohio, United States

Summary

Software Engineer with 2+ years of production experience building cloud infrastructure, deployment platforms, and automation at Siemens Digital Industries Software. Currently pursuing an M.S. in Computer Science at the University of Cincinnati, where I'm researching LLM-powered software engineering and AI systems. My work spans production cloud platforms, distributed systems, Kubernetes, CI/CD, backend development, and applied AI. Previously at Siemens, I:- Modernized deployment infrastructure using Kubernetes, AWS, Ansible, and GitLab CI/CD. - Executed 44 production upgrades while maintaining service reliability. - Reduced deployment time by 40% through Python automation and cut upgrade downtime by 50% using Blue-Green deployments.- Improved engineering quality by increasing automated code coverage from 35% to 78%. I'm interested in building scalable software across: • Backend Engineering • Cloud Infrastructure & Platform Engineering • Distributed Systems • AI/ML & LLM Applications Tech: Python • Java • Go • AWS • Kubernetes • Docker • Terraform • React • FastAPI • PyTorch • LangChain • SQL I'm always open to connecting with engineers, recruiters, and teams building large-scale software and AI-powered products.

Experience

University of Cincinnati
Graduate Researcher
July 2025 - Present (1 year 5 months)
Cincinnati, Ohio, United States

- Researching memory-augmented LLM agents for automated Android bug reproduction from user-recorded videos, focusing on AI-assisted software testing and mobile systems.
- Designed a two-stage multimodal pipeline that decouples video understanding from device automation using Google Gemini 2.5 Pro, Vertex AI, ADB, and UIAutomator2.

- Built an evaluation framework across 20+ Android applications, demonstrating 25.7× lower inference latency compared with the ViBR baseline while maintaining reproducible execution.
- Investigating multimodal reasoning, AI agents, GUI automation, memory systems, and scalable software engineering for next-generation automated mobile debugging.

Siemens Digital Industries Software

2 years 4 months

Software Engineer

July 2022 - June 2024 (2 years)

Pune District, Maharashtra, India

- Implemented a blue-green deployment strategy using Ansible automation, significantly reducing upgrade downtime by 75%.
- Managed data migration from AWS EFS, RDS, and Siemens DSS Services, ensuring robust disaster recovery capabilities through timely backups and a scripted restore process.
- Provided ongoing support and hotfixes to over 45 customer environments, guaranteeing seamless operations.

- Led a successful migration of a large-scale Teamcenter X environment from a traditional process-based architecture to a containerized infrastructure.
- Implemented GitOps, ArgoCD, and Kubernetes to streamline deployment and management,enhancing efficiency and scalability.
- Strengthened system security by integrating Hashicorp Vault for secure storage and management of sensitive data.

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- Led a successful project to implement Teamcenter X Cloud ART best practices across 8 teams.
- This initiative resulted in a 200% increase in code coverage for my team, significantly improving software quality and reliability.

Mentored and coached 4 new team members, leading to a 35% decrease in onboarding time.

Software Engineer

March 2022 - June 2022 (4 months)

Pune District, Maharashtra, India

- Gained proficiency in AWS basics like EC2, VPC, Security Groups, RDS, EFS, ECS, EKS
- Implemented Terraform for AWS infrastructure provisioning
- Executed Terragrunt tests on Terraform code. Learned importance of test cases in software development
- Studied Ansible and utilized ansible-playbook to install required packages on virtual machine and Run Terraform code with Terragrunt tests.

AISSMS College of Engineering
Undergraduate Researcher
January 2022 - May 2022 (5 months)

MediaRec: A Hybrid Media Recommender System

- Published MediaRec: A Hybrid Media Recommender System in IJRASET (2022), proposing a hybrid recommendation engine integrating content-based and collaborative filtering techniques.
- Designed and implemented recommendation algorithms including TF-IDF, K-Means Clustering, KNN, Pearson Correlation, and Cosine Similarity for personalized recommendations across movies, books, and music.
- Developed the end-to-end web application using Python, Flask, React, and Spotify API, translating machine learning research into a production-ready prototype.
- Collaborated with a faculty advisor and a multidisciplinary student team throughout the research, implementation, and publication process.

Education

University of Cincinnati
Master of Science - MS (Thesis), Computer Science · (August 2024 - December 2026)

Savitribai Phule Pune University
Bachelor of Engineering - BE, Computer Engineering · (2018 - 2022)